Sahaj Bhadja

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Summary

Forward Deployed Engineer with 2+ years shipping production **Python** and **LLM** tools, **RAG**, data pipelines, and workflow automation that delivered measurable outcomes, working directly with business stakeholders and payment and billing systems.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Advanced NLP, Database Systems, Analysis of Algorithms, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Data Retrieval, Software Engineering, Object-Oriented Programming, Operating Systems

Experience

USC Dornsife **September 2025 – Present**
Forward Deployed Engineer *Los Angeles, CA*

- Built fault-tolerant **LLM orchestration** for a deployed AI education platform with FastAPI workers, explicit job states, and failure isolation, processing 1K+ AI jobs at 3x throughput in production.
- Built **Python data pipelines** with incremental loads and reconciliation checks across SQL Server, PostgreSQL, and cloud storage, sustaining 99%+ reliability over 5K+ monthly jobs and cutting dashboard latency to 350ms.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Shipped an **LLM-powered workflow** end to end that replaced manual grading with prompt chains for rubric generation, vision OCR, and scoring, cutting feedback turnaround 60% for 150+ instructors.
- Orchestrated GPT-4o, Claude, and Qwen-VL behind a switchable provider layer with fallbacks and per-call **cost tracking**, handling 10K+ daily production requests on AWS at 90%+ grading consistency.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- **Automated manual workflows** with cross-platform data-sharing microservices that cut manual data entry 35%, and redesigned AWS Glue ETL that boosted pipeline efficiency 68% while cutting storage costs 15%.
- Partnered with Microsoft and GitHub stakeholders in Agile sprints from requirements to deployment, optimizing APIs to improve response times 30% and backing releases with 20+ **automated test** suites.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built a microservices platform for **billing, invoices, payments**, customer records, and lead management with REST APIs and real-time WebSocket updates, syncing 2M+ records across locations in production.
- Used analytics on feature usage and drop-off points to guide fixes that delivered **measurable business outcomes**, cutting resource wastage 42% and lifting lead conversions 15% through Jenkins CI/CD releases.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built multi-currency **payment** infrastructure with card, wallet, and PayPal flows plus idempotent webhooks for refunds, disputes, and chargebacks, cutting manual refund handling about 70% across 50K+ booking records.
- Built hourly Spark micro-batch pipelines on Databricks summarizing revenue, active users, and demand for internal dashboards, and **presented the end-to-end architecture** directly to the client before implementation.

Skills

Languages & Data: Python, SQL, TypeScript, Java, PostgreSQL, SQL Server, Azure SQL
AI & LLM: LLM Systems, RAG, Prompt Engineering, Embeddings, FAISS, Pinecone, OCR, GPT-4o, Claude
Data Engineering: ETL Pipelines, Spark, Databricks, AWS Glue, Data Validation, Reconciliation, Analytics, Tableau
Backend & Cloud: FastAPI, Flask, REST APIs, Microservices, AWS, Azure, Docker, CI/CD
Domain & Practices: Payments, Billing Workflows, Workflow Automation, Stakeholder Communication, Agile, Git, PyTest, Jenkins

Projects

Credit Document Analysis Assistant | *Python, FastAPI, FAISS, LLM APIs, PostgreSQL*

- Designed a **RAG** assistant over credit agreements and investment memos that answers questions with page citations, extracts covenants and key terms into structured tables, and tracks answer accuracy against an **evaluation set**.

ALM Reporting Automation Agent | *Python, SQL, LLM Agents, PostgreSQL, Tableau*

- Designed an **agent** workflow that pulls asset cash flows and liability projections, reconciles them across sources, computes duration gaps, and drafts monthly **ALM** commentary for analyst review before publishing.

Big-Brain Live | *LLaMA-2, FAISS, Pinecone, PostgreSQL, GCP, REST APIs*

- Built a **retrieval-augmented** document platform with LLaMA-2, transformer embeddings, and FAISS over 50K+ documents, reaching 95% query accuracy and cutting document review time 60% for 200+ users.